Practice Tests 1

Course Name: Database Principle Examination Time: 120minutes
Student Number:___________________Student Name:___________________

1. What is database integrity? How is it different from database security? (5+5)

2. What is the basic principle of database recovery? What is the implementation of database recovery? (4 + 6)

3. The relations R and S are shown in the figure below. Please provide the results of the following expressions.（5+10）

R

A	B	C	D
a_1	b_1	c_1	d_1
a_1	b_1	c_2	d_2
a_1	b_1	c_1	d_3
a_2	b_2	c_1	d_1
a_2	b_2	c_2	d_2
a_3	b_3	c_1	d_1

S

C	D
c_1	d_1
c_2	d_2

(1) Calculate the relational algebra operation: $\Pi_{A,C,D}(_{A!=a1}(R))\div S$
(2) Express the division operation using domain relational calculus.

4. Please answer the following questions:

With the coming of the World Cup, how to manage football activities has become a hot topic. In order to solve this problem, we built a Football Activity Management System, which needs to maintain the information of the Team, Player, Chief Coach, Chief Referee, Sponsor, And each Match. The data schema is set as follows:

[1] Team (TeamId, TeamName, PlayersNum, EstablishDate, ChiefCoach) ;
[2] Player (PlayerId,PlayerName, BirthDate, Height, Weight) ;
[3] Chief Coach (CoachId,CoachName, BirthDate, CoachLevel) ;
[4] Chief Referee (RefereeId,RefereeName, BirthDate, RefereeLevel) ;
[5] Sponsor (SponsorId, SponsorName)
[6] Match (MatchId, HomeTeamId, VisitTeamId, ChiefRefereeID, MatchDate, Score).

Assume that the football system meet the following arrangements:

[1] Each team has only one chief coach and several players.
[2] Each chief coach can only be hired by one team, and each player only play for one team.
[3] There are two teams in each match, one for the home team and one for the visiting team. There is only one chief referee for each match. Each match has a unique id and the date and score of the match are recorded.
[4] Each sponsor can sponsor multiple teams, but each team can only have one sponsor. Each sponsor can individually sponsor multiple players, and each player can also represent multiple sponsors.

[5] Additionally, it allows the cases where one team plays more than one matches in the same day or plays with the same team in the same day.

(1) Please draw the ER diagram of the football activity management system according to the above Settings (5).

(2) Please write Primary Key,Candidate Key and Foreign Key of the Match table(10).

5. Please answer the following questions:

The relationship between Teachers and Courses is shown in the following table:

Course ID(key)	Teacher Name	Department ID
C1	David	D1
C2	Mally	D2
C3	Musk	D1
C4	Musk	D1

(1) Which paradigm does the above relationship belong to? Why? (5)

(2) Does there exist the problem of "Delete Exception" or "Modify Exception"? If it does, how do they happen? (5)

6. Please answer the following questions:

It is known that the data relationship of local enterprises is as follows:

[1] Information of Employees: EMP(Eno, Ename, Sex, Age, Title)

[2] Information of Companys：CPY(Cno, Cname, Area)

Please answer the question according to the above information:

(1) Please CREATE an information table WORKIN where employees work in the enterprise and meet the following conditions: (10)

[1] The WORKIN table contains three attribute (the Eno from Emp, the Cno from CPY and the Salary)

[2] Define the primary and foreign key.

[3] Set a constraint that the Salary should not be less than 3,000 yuan.

(2) Based on the WORKIN table created in question (1), fill in the following SQL statement implementation function:

a) Query the Cno and Cname of the company which has the most employees (2+3)

```
SELECT C.Cno,C.Cname
FROM Company AS C， WORKIN AS W
WHERE C.Cno = W.Cno
GROUP BY ____(A)____
HAVING ____(B)____  （SELECT COUNT(*)
                     FROM WORKIN AS W2
                     GROUP BY W2.Cno）
```

b) Query the Cno and Area of the Company whose employees have the oldest average Age (2+3)

```
SELECT C.Cno， C.Area
FROM EMP AS E， CPY AS C， WORKIN AS W
WHERE W.Cno = C.Cno
AND W.Eno = E.Eno
        GROUP BY C.Cno, C.Area
HAVING ____(C)____ >= ____(D)____
```

(3) Please fill in the vacancy part of the following TRIGGER to realize the automatic maintenance function of employee salary. When the Title of the employee changes, the salary of the new title should be calculated according to the "Title-Salary" Function——SalaryList(Title) . And the salary should be updated automatically: (1+2+2)

```
CREATE TRIGGER NewSalary
AFTER UPDATE ____(E)____ ON EMP
REFERENCING NEW AS N
FOR EACH ROW
BEGIN
        UPDATE WORKIN
        SET ____(F)____
        WHERE ____(G)____
END
```

7. Please answer the following questions:

Suppose there is a Electronic Toll Collection (ETC) related to the highway:

When a vehicle drives out of the station, firstly, we read the current balance X from its credit card A, and we records it as X = R(A); Then we deduct the cost S and we records it as X = X - S; Finally, we write the balance X into credit card A, and we records it as W(A,X). We also assume that more than one vehicles can use the same credit card at the same time.

Please answer the following questions based on the above assumptions:

(1) If two vehicles T1 and T2 bound to the same credit card A drive out of the cost station at the same time, their order execution sequence is as follows:

T1: X1 = R(A),X1 = X1 - S1, W(A,X1),

T2: X2 = R(A),X2 = X2 - S2, W(A,X2),

What problems will occur at this time? Could you explain why? (5)

(2) To solve the above problems, the exclusive Lock XLock(A) is introduced to lock data A, and the UnLock UnLock(A) is introduced to unlock data A. Please supplement the above execution sequence to make it meet the 2PL protocol. (5)

(3) When two vehicles bound to the same credit card number pass through the toll booth at the same time, the possible sequence of instruction executions is: x1=R(A), x1 =x1-a1, x2 = R(A), x2 = x2-a2, W(A， x1), W(A， x2). To solve the above problem, the exclusive lock instruction XLock(A) is introduced to lock the data A, and the unlock instruction Unlock(A) is used to unlock the data A. Please supplement the above execution sequence to satisfy the 2PL protocol.(5)